

THEJUS MAHAJAN PH.D.

Curriculum Vitae

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*30.09.1992 (Talikulam, India)

Married

Nationality: Indian



PROFILE

- Computational Scientist (PhD, Post-doc) with experience in data pipeline engineering, mathematical modelling, and high-performance computing.
- Core skills: R (tidyverse), Python (JAX, Pandas), Fortran, SQL, Git.
- Currently: Internship in clinical data research at HealthTwist GmbH (medical data pipeline, 143K+ records).
- Available from mid-April 2026** (earlier by arrangement).
- Open to: Data Science, Biostatistics, Bioinformatics, Clinical Data Analysis, Research Software Engineering.

PROFESSIONAL EXPERIENCE

Internship – Clinical Biostatistics & Data Pipeline Engineering HealthTwist GmbH

02/2026 - 04/2026 Berlin, Germany

Clinical data research and development under Dr. Andreas Busjahn

- Refactoring a monolithic 1,834-line R data pipeline (tidyverse) for Germany's interventional radiology quality registry (DeGIR); 143,000+ patient records.
- Externalisation of 257 hard-coded correction rules into 8 configuration CSV files; 26.5% code reduction (1,834 → 1,348 lines).
- Development of a parameterised complication grading function replacing 124 lines of repetitive code.
- Replacement of 72 deprecated function calls with modern tidyverse equivalents (gather/spread → pivot_longer/pivot_wider). All changes byte-identical verified via identical() comparison.

Continuing Education – Bioinformatics and Biostatistics CQ Beratung + Bildung GmbH

08/2025 - 02/2026 Berlin, Germany

Applied bioinformatics and biostatistics

- Programming methodology: Shell (BASH), Python (object-oriented), R.
- Bioinformatics databases (NCBI, Biopython, SQL) and sequence analysis.
- NGS analysis pipelines (Next-Generation Sequencing; Nextflow, Galaxy); structural bioinformatics.
- Biostatistics (R/Bioconductor): ANOVA, PCA, HCA, multivariate analysis, Markov chains.

Guest Scientist (parallel to continuing education from 08/2025) Helmholtz-Zentrum Hereon, Ecosystem Modelling

05/2025 - 10/2025 Geesthacht, Germany

Modelling evolutionary processes in marine ecosystems; simulation of long-term environmental impacts using advanced computational methods.

Job Search & German Language Course

02/2025 - 04/2025 Hamburg, Germany

Active job search and German language acquisition (Goethe-Zertifikat B1).

KEY STRENGTHS



Data Pipeline Engineering

Refactoring a production data pipeline (1,834 lines, 143K records) with 26.5% code reduction at zero behavioural regression. Byte-level verification after every change.



Complex Problem Solving

Resolved critical data loss by building a Fortran simulation to correct a misaligned particle detector; recalculated results using multi-sensor data.



Bioinformatics & Biostatistics

Bridging physics expertise with modern biological workflows. Proficient in NGS data analysis, sequence alignment, and multivariate statistical modelling with R (Bioconductor) and Python.

TECHNICAL SKILLS

Programming

R (tidyverse, dplyr, tidyr, stringr, readr, R Markdown), Python (JAX, Pandas, NumPy), Fortran, C++, SQL, BASH

Data Engineering

Pipeline refactoring, config-driven architecture, identical() verification, large datasets (HDF5, NetCDF)

Bioinformatics

NGS analysis, Nextflow (DSL2), sequence analysis, Bioconductor

Biostatistics

Multivariate analysis, PCA, ANOVA, dimensionality reduction, tidymodels

Tools

Git/GitHub (branching, collaborative workflows), Linux/Unix, HPC clusters, Conda, LaTeX

LANGUAGES

English



Fluent (C1)

German



B1 (Goethe Certificate), B2 in preparation

French



Basic knowledge

Malayalam



Native

Post-doctoral Scientist

Universität Hamburg, Institute for Marine Ecosystem and Fishery Science

08/2021 - 01/2025 Hamburg, Germany

Marine ecosystem modelling

- Parental leave from 01/2024 to 12/2024.
- Model development: Development of the "Cyanobacteria Life Cycle" (CLC) model within the ERGOM framework for climate warming projections.
- High-performance computing (HPC): Translation of a legacy Fortran model into a high-performance Python engine (Google JAX, TPU/GPU parallelisation). Processing of large NetCDF datasets in Linux HPC environment.

Career Transition

04/2021 - 07/2021 Hamburg, Germany

Relocation from India to Germany and active job search.

Physics Tutor & Chess Coach

Freelance / Tutorwaves Solutions Inc (from 08/2020)

10/2018 - 03/2021 Kerala, India

Freelance physics tutoring and coaching of state-level chess teams; from 08/2020 at Tutorwaves: creation of physics content and mentoring of students in the USA and Australia.

Doctoral Researcher in Astrochemistry

Université Paris-Saclay

10/2015 - 09/2018 Orsay, France

Conducting atomic collision experiments, data analysis, and results modelling.

- Acquisition and processing of terabytes of raw data from tandem particle accelerator experiments.
- C++ code optimisation: improvement of existing algorithms for noise suppression, signal smoothing, and conversion of raw data into multi-dimensional structures.
- Development of a molecular fragmentation model for predicting chemical reactions in space (contribution to the KIDA database).

EDUCATION

Ph.D. in Astrochemistry

Université Paris-Saclay, Institute of Molecular Sciences (ISMO)

10/2015 - 09/2018 Orsay, France

Supervisor: Dr. Karine Béroff. Thesis: *Excitation and fragmentation of C_nN^+ ($n=1-3$) in collisions with He atoms at intermediate velocity.*

Research Project – Theoretical Atomic Physics

IIT Mandi (under Dr. Hari Varma)

02/2015 - 08/2015 Mandi, India

Theoretical atomic physics; partly remote.

M.Sc. in Physics

National Institute of Technology Calicut, India

07/2012 - 12/2014 Calicut, India

First Class with Distinction (CGPA 8.71/10).

B.Sc. in Physics

University of Calicut, India

06/2009 - 04/2012 Thrissur, India

Grade: B+ (CGPA 3.49/4.0).

Hamburg, 11 March 2026

Dr. Thejus Mahajan

Tamil



Fluent

Hindi



Basic knowledge

COURSES & TRAINING

High Performance Computing (HPC) Training

Jülich Supercomputing Center (JSC),
Forschungszentrum Jülich

MITx: 7.00x Introduction to Biology – The Secret of Life

Comprehensive coursework in genetics, biochemistry, and molecular biology.

PUBLICATIONS

T. Mahajan et al. *J. Phys.: Conf. Ser.*, 1412:142026, 2020.

T. Idbarkach et al. *Astron. Astrophys.*, 628:A75, 2019.

T. Mahajan et al. *J. Phys. B*, 2019.

T. Idbarkach et al. *J. Phys. B*, 51(24):245201, 2018.

T. Idbarkach et al. *J. Phys.: Conf. Ser.*, 1412(11):112028, 2020.

Google Scholar Profile